Operative Techniques in
Orthopaedics

Prosthesis Options and Management in Upper Extremity Amputation

Christopher Fink,[†] and Yoni Diamond[‡]

Amputation surgery is a life-altering experience, profoundly impacting an individual's ability to perform daily tasks and engage in their chosen activities. Upper extremity prostheses have advanced remarkably, offering diverse solutions to restore functionality and enhance the quality of life for amputees. This article provides a comprehensive examination of the various types of upper extremity prostheses available today, highlighting their unique features, advantages, and limitations. This article categorizes upper extremity prostheses into three main types: body-powered prostheses, myoelectric prostheses, and advanced prosthetic technologies. Passive prostheses are generally designed to replicate the appearance of the missing limb. Body-powered prostheses rely on cables and harnesses for control, offering affordability and durability. Myoelectric prostheses employ electromyography (EMG) signals from residual muscles for intuitive and precise control. Advanced prosthetic technologies, including neural interfaces and 3D printing, are pushing the boundaries of what is possible in terms of functionality and customization. This article aims to serve as a valuable resource for upper extremity surgeons to help guide options and even optimize surgical plans to allow individuals to maximize the use of upper extremity prostheses.
Oper Tech Orthop 33:101061

KEYWORDS upper extremity prosthetics, upper limb amputation, partial hand prosthesis, partial hand amputation, body powered prosthesis, external powered prosthesis, passive prosthesis, hybrid prosthesis, activity specific prosthesis, myoelectric prosthesis

Introduction

Technological progression continues to expand options for upper limb prostheses with every passing year. Whereas in the past one might have been confined to a few options, prostheses now have a wide array of options, applications, and complexity. The consideration of which type of prosthetic device depends on several variables including length and quality of residual limb, vocational and avocational demands, goals, strength, range of motion, comorbidities, cognitive ability of the patient, insurance coverage, and aesthetics. Upper limb function is extremely complex, and the vast array of tasks any 1 individual may perform may require multiple prostheses for optimal function.

[†]Handspring Clinical Services, Middletown, NY.
[‡]Hanger Clinic, Brooklyn, NY.
Address reprint requests to Chris Fink, MSPO, CPO, Handspring Clinical Services, 4 Riverside Dr, Middletown, NY 10941. E-mail: chris.fink@myhandspring.com

The most common level of upper limb amputation is at the partial hand (PH) level with finger amputations being the most common.[1] There is an important balance between preserving length and improving functional outcomes with a prosthesis. In certain instances, excessive length can limit prosthetic options by reducing space for components. It is therefore important to consider the desired prosthetic design and components when performing amputations.[2]

For the purposes of this article, prosthetic options will be divided into 7 categories which are shown in Table.

Not using a prosthesis is always a consideration either because it is too difficult or painful to use for the patient or they simply do not find the need to use one.

Passive restorative prostheses are designed to replicate the missing part of the body. While this type of prosthesis can be used at any level it is often applied at the PH level and has the benefit of protecting the residual limb, providing mild compression which can help with pain, and improved psychosocial function. These prostheses are immobile and

https://doi.org/10.1016/j.oto.2023.101061
1048-6666/

Table List of the 7 prosthetic options and examples of each option.

Prosthetic Option	Example
No Prosthesis	Patients opts to not use prosthesis
Passive Restorative	Silicone aesthetic hand
Passive Mechanical	Ratcheting mechanical finger
Body-Powered	Transradial prosthesis with hook and harness
External Powered	Transradial prosthesis with myoelectric control and electric hand
Hybrid	Transhumeral prosthesis with body-powered elbow and electric hand
Activity Specific	Transradial prosthesis with bicycling terminal device

therefore can still provide a stable surface for opposition, as well as light grasp, pushing, and pulling.

Passive mechanical devices rely on a positional mechanism such as a ratcheting lock to place the prosthesis in various positions. A patient can take the prosthetic finger(s) and bend them around an object, for example, with their sound hand. They then need to release the digits again with their sound hand. Such devices are useful in that they are strong and stable once set into the desired position and do not require the patient to use energy to hold the desired position. This would be particularly useful for a patient working in more strenuous situations such as holding heavy bags, construction, or yard work.

Body-powered prostheses rely on the patient's motion and energy to operate the device. Common motions used are glenohumeral flexion and bi-scapular abduction. This energy and motion are captured through the harnessing system that runs around the user's contralateral limb and to the terminal device allowing it to be open and closed. Such systems are durable, easier to repair than electronic devices, allow for the patient to control how much force they are applying, and are less costly. Contraindications may be cognitive limitations, a very short residual limb or limited range of motion that cannot provide adequate excursion to operate the prosthesis, and/or inadequate strength.

External-powered prostheses rely on electric power to drive motors to operate prosthetic componentry such as multi-articulating hands or elbows. Various control methods can be used to operate externally powered prostheses, and these are discussed later. Advantages include more complex operational capacities such as various grip patterns compared to body-powered terminal devices, increased field of use as users are not limited by a harness and cable system, less energy consumption, and decreased reliance on the sound side. In addition, users who have a limited range of motion and cannot provide adequate cable excursion may be a better candidate for external power. Disadvantages include an increase in weight, an increase in complexity of control, decreased durability, and increased cost.

Hybrid prostheses simply incorporate various components from these different categories of prostheses. For example, an SD patient may benefit from a multi-articulating electric hand, but it is not desirable to add excess weight to the prosthesis, so the shoulder and elbow joints could be body-powered. This can also help reduce the cognitive load of operations. If a patient needs to rely on myoelectric control of the shoulder, elbow, wrist, and hand it may become difficult and overwhelming.

Activity-specific prostheses encompass a group that is designed to achieve a specific function. For example, a WD patient may have a prosthesis with an attachment specifically for lifting weights at the gym or riding a bicycle. This is particularly useful to avoid the breakdown of primary prosthesis or for use only during specific activities. For example, a congenital patient with a limb difference may be able to accomplish most ADLs without a prosthesis but have use for 1 designed to hold a pen or play a musical instrument.

Partial Hand Prostheses

Options for partial hand prostheses have increased greatly over the last decade, and depend on the level of amputation. Like more proximal amputation levels, partial hand prostheses are available in passive, body-powered, and external-powered designs.

Passive restorative finger or hand prostheses are typically made from silicone and hand painted and sculpted and can improve 2-handed grasp, light push/ pull activities, protect the residual limb, and improve psychosocial function.

Passive mechanical design (Fig. 1) devices mechanically lock into distinct positions and function is reliant on the contralateral limb or surrounding environment for positioning of the digits.

Body-powered partial hand prostheses (Fig. 2) rely on the motion of more proximal joints to control the motion of the prosthetic device and provide grip strength proportional to the strength of the input provided by the user.

External powered partial hand prostheses utilize motors, processors, and batteries to open and close the prosthetic fingers.

Terminal Devices

The most distal component of the prosthesis is referred to as the terminal device. No single terminal device can replicate

Figure 1 Passive mechanical fingers. Images courtesy of Naked Prosthetics. (Color version of figure is available online.)

Figure 2 Body-powered fingers. Images courtesy of Naked Prosthetics and Partial Hand Solutions LLC. (Color version of figure is available online.)

human hand function completely, thus multiple prosthetic terminal devices may be recommended.

Passive restorative terminal devices are sculpted and painted to match the contralateral limb as closely as possible. Fabrication varies from semi-rigid foam, silicone, and wire to mechanical jointed digits which can allow for stabilizing, pushing, pulling, balancing, and gross 2-handed grasp. There may be additional psychosocial benefits by improving social acceptance and reducing attention brought to the wearer's limb difference.

Activity-specific terminal devices (Fig. 3) can be interchangeable and may allow users to return to previous recreational activities or hobbies that they thought may not be possible.

Body-powered terminal devices (Fig. 4) capture more proximal joint motions via harness system and transmit the motion to the terminal device through cables.

External powered terminal devices (Fig. 5) (sometimes incorrectly referred to as myoelectric terminal devices) utilize motors, processors, and batteries to open and close. Control of these devices is provided by surface electrodes placed inside the prosthetic socket and against the user's limb. While electrodes are most common, alternative inputs like switches, touch pads, and linear transducers are also available.

Single-articulating hands have 1 degree of freedom and traditionally use the 3-jaw chuck/ tripod pinch position in the hand. Their simple design and single motor tend to be faster and provide stronger grip force compared to multiarticulating hands.

Multiarticulating hands (Fig. 6) have multiple degrees of freedom and have individual motors that control each finger separately. This allows for the prosthetic fingers to close around complex shapes and allows for multiple grasp patterns to be made by the prosthetic hand.

Wrist Units

The wrist unit refers to the mechanism which connects the terminal device to the forearm of the prosthesis. They can provide supination/ pronation, ulnar/ radial deviation, and flexion/ extension and can decrease the compensatory motions required for the users to complete tasks.

We will divide these into nonpowered and powered wrist units.

Nonpowered wrist units are used in passive, body-powered, and activity-specific prostheses (Fig. 7) and have various methods of tightening loosening, or placing the hand in space.

Powered wrist units are used in externally powered and hybrid prostheses. The simplest powered wrist is the *electronic quick disconnect (EQD)*. This wrist allows for the control signals to pass from the inputs (ie, electrodes, touch pads, switches) to the terminal device.

Figure 3 Activity-specific terminal devices. Images courtesy of Fillauer TRS. (Color version of figure is available online.)

Figure 4 Body-powered hooks and hand. Images courtesy of Fillauer. (Color version of figure is available online.)

Figure 5 External powered hook and single articulating hand. Images courtesy of Fillauer Motion Control. (Color version of figure is available online.)

Powered wrist rotators provide the same features as the EQD but with the addition of the inputs being able to actively control supination/ pronation (Fig. 8) and can improve the speed of completing tasks, reduce compensatory motions, and eliminate the need for contralateral limb involvement.

Passive wrist flexion can allow the terminal device to lock into different flexed/ extended positions, or radial/ ulnar deviations (Fig. 9), while *powered wrist flexion* allows for wrist flexion/ extension positions to be controlled through the same or different inputs that are used for terminal device control (Fig. 10).

Elbows Joints

Prosthetic elbow joints can be passive, body-powered, externally powered or hybrid in design (Fig. 11). There are multiple types of elbow joints within each category with each one having pros and cons related to its function and the requirements of the user. Matching the appropriate elbow type to the user and their needs is crucial for overall success.

Shoulder Joints

The array of prosthetic shoulder joints is similar to other joints, however (Fig. 12) unlike other prosthetic joints, motion is not controlled by body motions or motors. In this case, body-powered and external-powered refers to the locking mechanism of the shoulder joint. Motion is controlled by the contralateral limb or by using body position and gravity. In body-powered shoulders, the joint locks and unlocks through tension on a cable, which is commonly operated using a manual switch or lever. Passive shoulder joints rely on friction for positioning.

Control Methods for Body-Powered Prostheses

Voluntary-open (VO) terminal devices are closed at rest and when tension is applied to the cable the TD opens. This allows for *passive* holding of an object. Grip force is determined by the amount of rubber bands or springs on the device (Fig. 13).

Alternatively, voluntary-close (VC) terminal devices are typically seen on transradial level body-powered prostheses, are open at rest, and grip force is controlled by the user and is only limited by the user's strength and ROM.

Control Methods for External Powered Prostheses

While myoelectrics continue to be the most common method for control of external-powered prostheses it is crucial to understand that additional inputs are available for more proximal-level amputations or unique anatomical presentations.

Myoelectric control refers to the use of electrodes to detect electromyography (EMG) signals in the residual limb to

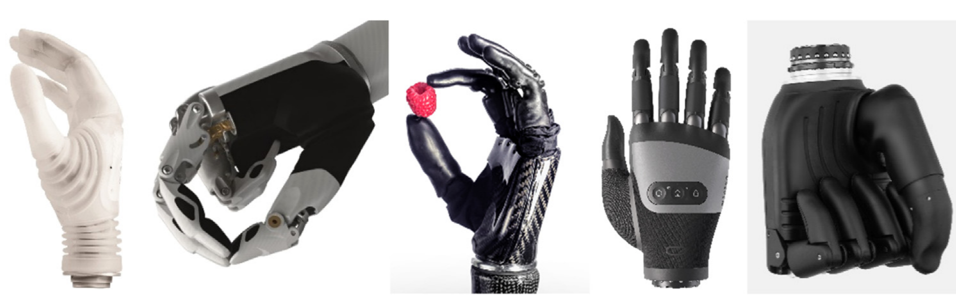

Figure 6 Multiarticulating hands, various designs. Images courtesy of OttoBock, Psyonic, Fillauer Motion Control, Aether Biomedical. (Color version of figure is available online.)

Figure 7 Nonpowered wrist units, friction, flexion, and 4-function. Images courtesy of Fillauer. (Color version of figure is available online.)

Figure 8 Powered wrist rotator. Image courtesy of Fillauer Motion Control.

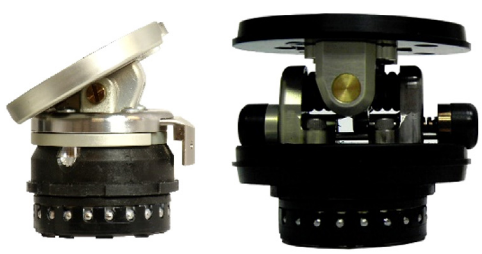

Figure 9 Flexion and multi-flex wrist units. Image courtesy of Fillauer Motion Control. (Color version of figure is available online.)

Figure 10 Powered wrist flexion unit. Image courtesy of Fillauer Motion Control. (Color version of figure is available online.)

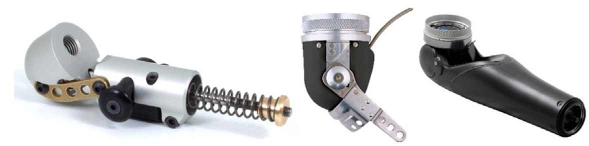

Figure 11 Variety of prosthetic elbow joints. Images courtesy of Fillauer, Fillauer Motion Control. (Color version of figure is available online.)

provide inputs for prosthetic control. *Single-site* or *dual-site* control has been traditionally used for years. In these control systems, 1 or 2 electrodes are placed over appropriate muscle groups or myo-sites. These sites are traditionally placed on large flexor and extensor muscle groups in the segment just proximal to the amputation (ie, wrist flexors/ extensors in transradial amputations and elbow flexors/ extensors in transhumeral amputations). For dual-site control to work optimally 2 *strong and separate* EMG signals need to be detected. This is done with a myo-testing device. Single-site control is used when only 1 strong and separate signal is available. To control multiple prosthetic joints (ie, terminal device, wrist, elbow) a switch mechanism needs to be included.

Multisite control allows for multiple joints to be controlled with different muscle groups. For example, 2 muscles can control hand open and close while 2 other muscles control wrist rotation. Due to limited surface area on most limbs and crosstalk (electrodes detecting nearby EMG signals that are

Figure 12 Mechanical prosthetic shoulder joint. Image courtesy of FIllauer. (Color version of figure is available online.)

Figure 13 Traditional rubber band VO device and variable grip force device (JAWS). Images courtesy of Fillauer and Fillauer TRS Inc. (Color version of figure is available online.)

Figure 14 Alternate inputs for external powered prostheses. Images courtesy of Fillauer Motion Control. (Color version of figure is available online.)

not desired), this control system can be difficult to implement.

Pattern recognition is a more recent option for myoelectric control and employs multiple surface electrodes that synthesize all the EMG signals being produced at 1 time. This means that weaker signals and crosstalk are no longer issues. They are programmed by having the user make various motions in their residual limb to control the desired motions of the prosthesis. For example, if a user wanted to control terminal device open/ close as well as wrist pronate/ supinate, 4 different "patterns" of EMG signals would need to be detected. Pattern recognition allows for control of multiple grip patterns in a muti-articulation hand as well as multiple joints without the need for a switch. It can even be argued that this type of control is more intuitive for the user, especially when combined with surgical techniques like targeted muscle reinnervation.

Alternative input control strategies (Fig. 14) are traditionally used on more proximal level amputations, or where traditional myoelectrics are unable to be used. This allows the user to still experience the benefits of externally powered prostheses even though they may not have sufficient EMG signals to control the device. These may include touch pads, linear transducers or potentiometers, and a variety of switches.

The Importance of the Prosthetic Treatment Plan

While prosthetic terminal devices, wrists, elbows, and control mechanisms are significant factors in the success of a prosthesis, the most crucial factor continues to be socket fit. Without a well-fitting socket, the componentry placed on it is immaterial. Therefore, the skills and talents of the prosthetist collaborating with these users are paramount to the user's ability to achieve their goals.

Secondary to creating a successful socket, is executing an appropriate control mechanism. This can be achieved through optimizing harnessing and cabling. It can also include determining which prosthetic technology to use, and more importantly, which technology not to use. Each device from partial hand to shoulder level and passive to external powered have their pros and cons. Matching the user with the correct types of prosthetic technology is just as important as providing a well-fitting socket and reliable control mechanism. With the options expanding at an unprecedented rate, this ability to match the user with the most appropriate devices is irreplaceable.

Declaration of Interest Statement

The authors declare no competing interests.

References

1. Braza DW, Yacub Martin JN: Upper limb amputations. In: Frontera WR, Silver JK, Rizzo TD (eds): Essentials of Physical Medicine and Rehabilitation, ed 4, Philadelphia, PA: Elsevier, 651-657, 2020
2. Miguelez J, Conyers D, Lang M, et al: Upper extremity prosthetics. In: Lenhart MK (ed): Care of the Combat Amputee, Washington, DC: Office of the Surgeon General, Department of the Army, United States of America, 607-640, 2009